

Homo Faber in the making: A cross-cultural analysis of toolmaking skill acquisition in non-industrial societies

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1 Introduction

Toolmaking, together with bipedalism and language, was once regarded as a defining feature that makes humans distinct from other species (Oakley, 1949). It has also been constantly portrayed as the pinnacle in the evolution of cognition and creativity (e.g., Bergson, 1998; Vaesen, 2012). However, this simple notion of human uniqueness and its associated linear evolutionary narrative was overturned in the past decades by mounting evidence of toolmaking by non-human primates and corvids (Shew, 2017; Shumaker et al., 2024), particularly in the domain of food foraging. As pointed out sharply by Shumaker et al. (2024: 246-248), even some of the higher-order cases of tool manufacture, such as using existing tools to make other tools, using tools in combinations for a single task, and cooperation in making tools, have been demonstrated to be not uniquely human in an absence/presence manner. From a comparative perspective, it seems that the recruitment of toolmaking process for novel functions (Davidson & McGrew, 2005) has not yet been recorded in any species other than the human lineage. In terms of time allocation of tool-use and toolmaking as well as the level of dependency in caloric intake (Shipman, 2010), toolmaking occupies a central niche in our daily activities as compared with other toolmaking species.

On top of the basic survival need, it is also critical to human experience in the sense that a bio-cultural feedback loop is formed through toolmaking that shapes our minds and bodies in turn (Birch, 2021; Malafouris, 2012, 2021; Stout, 2021), as demonstrated by an extensive corpus of literature revolving around stone tools due to their longevity and great preservation. For example, a series of neuroimaging studies designed by Stout and colleagues (Stout & Chaminade, 2007; Stout & Hecht, 2023; Stout & Khreisheh, 2015) revealed the shared neuroanatomic foundations and differential cognitive demands of various lithic technologies, highlighting the evolutionary importance of perceptual-motor adaptation in technological evolution. Stone toolmaking and use has also been identified as a major selection force driving the diachronic changes in human hand morphology, including both bone and soft tissue anatomy, based on multiple biomechanical/kinematic studies in the context of experimental archaeology (Dunmore et al., 2023; Karakostis et al., 2021; Kivell et al., 2023; Marzke, 2013)

Yet the skill of toolmaking is not born with us — naive individuals need to acquire a given skill either through observing and learning from others (social learning) or repeated trials and errors (individual learning). Very often the skill attainment, particularly this type of motor skill involv-

ing extensive hand-eye coordination, requires a combination of these two learning strategies. Although this behavior is of great significance in human evolutionary studies as argued above, the mass machinery manufacturing process of tools introduced by industrialization weaken the necessity of handicraft practices and their inter-generational transmission in our contemporary daily life. Consequently, ethnographic narratives depicting the subsistence and economy of non-industrial societies, mostly written during the nineteenth and twentieth centuries, provide us a precious opportunity to investigate this topic, among which hunter-gatherer societies represent the most well-studied and well-published sample.

Embedded within the century-old anthropological research tradition revolving around childhood, the learning dynamics of hunter-gatherers have garnered considerable attention in the past two decades due to the burgeoning literature of cultural evolutionary theories ([Hewlett et al., 2024](#)). Garfield et al. ([Garfield et al., 2016](#)) conducted the first systematic investigation on hunter-gather social learning, involving 982 ethnographic texts from 23 different communities. In this research, they characterized the commonalities of learning behaviors across diverse domains regarding their transmission mode, forms of teaching, and learners' age. Within the acquisition of subsistence skills, which dominated the early and middle childhood, they concluded that oblique transmission becomes more important when compared with vertical transmission with the increasing age of learners. Subsequently, Lew-Levy and her colleagues narrowed down the learner' identities to children and thoroughly examined their learning patterns of subsistence skill, of which toolmaking skill is a subset, depicting a more detailed picture regarding the dominant learning processes and skill types across each developmental periods ([Lew-Levy et al., 2017](#); [Lew-Levy, Milks, et al., 2020](#)). Drawing upon both the cross-cultural analysis of existing ethnographies and new systematic field data collected from certain foraging communities, this children-centered research line of inquiry remained to be the most fruitful one in the past few years, emphasizing the evolutionary significance of object/toy as a medium of learning ([Lew-Levy et al., 2022](#); [Riede et al., 2023](#)) and the often ignored strategy of peer-learning ([Lew-Levy, Kissler, et al., 2020](#); [Lew-Levy et al., 2023](#); [Lew-Levy & Amir, 2024](#)). In addition, other researchers conducted either literature survey ([Bira & Hewlett, 2023](#)) or in-depth systematic observation ([Schniter et al., 2023](#)) on the knowledge transmission patterns of societies featuring other forms of subsistence strategies. For example, Bira and Hewlett ([2023](#)) compared the cultural learning patterns between hunter-gatherers and pastoralist children and found that peer-learning was less likely to be documented in the latter compared with the former context. Based on long-

term field data of Tsimane forager-farmers, Schniter et al. (2023) identified the importance of older same-sex relatives and multigenerational pedagogy in the successful reproduction of crucial skills like handicrafts.

Some pressing issues still remain unsolved despite previous intellectual efforts on the topic tool-making skill acquisition. First and foremost, the extremely low contribution of individual learning in subsistence skills, particularly toolmaking, which according to Garfield et al. (2016) only occupy 3.7% of all cases. There could be several reasons behind it. An obvious one is that ethnographers may systematically care more about communal activities and it's challenging to follow and observe individuals. The other source of bias could be coding. And that what I will try to improve in this project. I have a coding scheme that provide a more detailed evaluation of the learning processes. Second, the unbalanced representation of hunter-gather societies versus all other subsistence types within non-industrial societies mask our understanding because tool-making (by hand) is not unique to hunter-gatherers.

To understand the diversity and regularity in the process of human toolmaking skill acquisition, this project analyzed 170 non-industrial societies displaying varying subsistence strategies around the world using the eHRAF World Cultures database, with a particular focus on developmental contexts and transmission biases involved in the learning processes of toolmaking skills. At the most basic level, this study provides the most in-depth descriptive investigation on this topic so far (Gerring, 2012). It can help improving the external validity of archaeological experimental design. It can serve as a middle-range framework to better understand the formation of archaeological records. This study investigates the processes of tool-making transmission in non-industrial societies, focusing on fundamental aspects of social learning: who, where, when, how, what, and why. For the “who” aspect, I examined the identities of both the learner and the model, as well as their social relationships, which corresponds to the transmission mode question in cultural evolution. The “where” question addresses the learning context, such as whether it occurs indoors or outdoors. The “when” explores the ages of both the learner and the model. For the “how,” I investigated the degree of social interaction involved in the learning process. The “what” focuses on the content of transmission, such as pottery-making, woodworking, or other craft activities. Lastly, the “why” examines the motivations behind a learner's choice of model—whether it is based on the model's recognized expertise, convenience, or familial ties. Gender divisions and their influence on these dynamics are also considered under the “why”

question.

Built upon the answers to the six aforementioned questions, this study explores whether hunter-gatherer societies differ from other subsistence systems in these traits. By analyzing these variables, I aim to provide a systematic understanding of tool-making transmission and its variability across cultural contexts. Therefore, this paper intends to answer the following three research questions. 1) Do hunter-gatherers have a unique learning pattern compared with groups practicing other types of subsistence strategies? 2) Do learning strategies change across different developmental periods? (More specific hypothesis: will the frequency of social learning decrease through time?) 3) What are some major transmission biases involved? (kin-based? context(success)-based?)

Engaging with the works of Lancy ([Lancy, 2017](#))

2 Methods and materials

2.1 Searching strategy

This study analyzed a global sample composed of 170 non-industrial societies indexed in the eHRAF World Cultures database, which include 45 hunter-gathers, 36 horticulturalists, 29 other subsistence combined, 28 intensive agriculturists, 14 primarily hunter-gatherers, 13 agropastoralists, and 5 pastoralists cultures (**Figure 1**). Societies labeled as commercial economy are pre-excluded here since they are often associated with industrialization to some degree. Because the eHRAF database is organized using the Outline of Cultural Materials (OCM) coding scheme, the searching strategy focuses exclusively on one OCM code that is most relevant to the research question, namely OCM 868 “Transmission of Skills”. Under this OCM code, this project uses the following keywords for searching, including tool, make/made/making, produce/producing, manufacture/manufacturing, create/creating, construct/constructing, assemble/assembling, fabricate/fabricating, shape/shaping, forge/forging, form/forming. These keywords together yields 3060 paragraphs, among which approximately 30-40% are repetitive.

Focusing on the toolmaking skill acquisition in non-industrial societies, this project adopts the following exclusion criteria when screening the paragraphs yielded in initial search, including archaeological and historical literature sources, cases of purely tool use, instance where local

people taught the ethnographer how to make tools, instances where it was clearly stated a new form of government-initiated school to save a certain craft, instances where individuals are taught how to make Christianity-related artifacts such altar, and instances where indigenous people mentioned that they learn a specific craft from museum pieces or photographs. However, these cases may appear as contextual information in the appendix to provide more details of the technologies themselves. In the end, the eligible paragraphs and their associated contextual paragraphs were compiled as an 290-page and geographically organized appendix available as supplementary materials of this article.

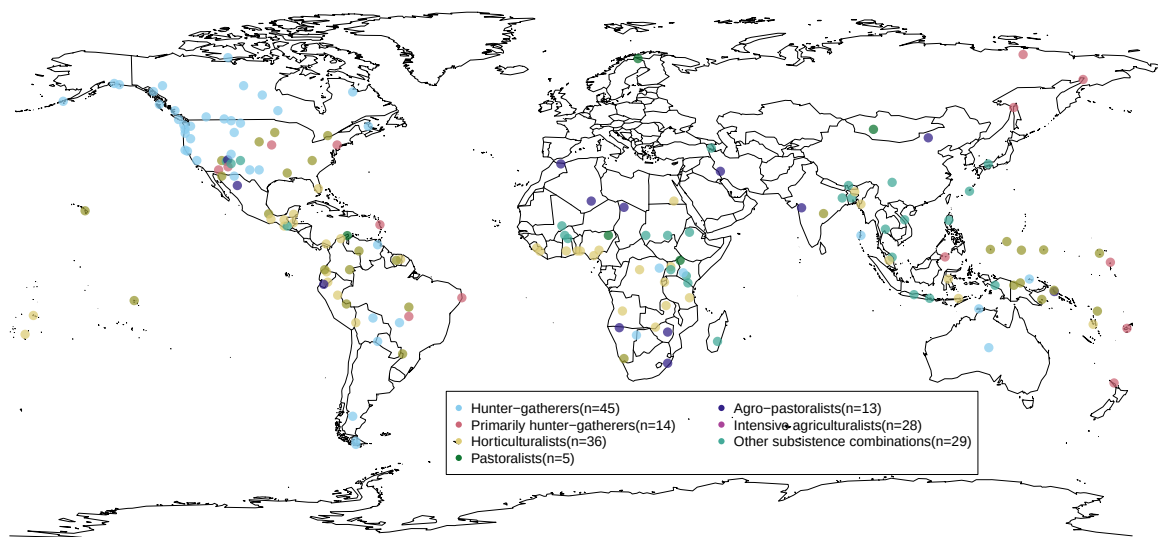


Figure 1: The spatial distribution of non-industrial societies covered in this study (n=170). It should be noted that these coordinate data represent language ‘centroids’ from the open-access language documentation site glottolog.org instead of the real geographical locations of given communities. These language centroids essentially represent the center point of the area where an relevant language is spoken.

2.2 Coding scheme

In the ethnographic coding, the acquisition of a specific toolmaking skill (e.g., clothes making) documented in one culture is regarded as an instance (i.e., a single row in the spreadsheet), which can be compiled based on multiple sources by different authors. To ensure the relia-

bility of coding results, the whole appendix was coded twice by the author until all discrepancies were resolved. The author employed a dual-system recording approach with some key variables. Specifically, the author first documented the descriptions provided by ethnographers and then categorized the information in a separate column for quantitative analysis. It should be noted that “NA” is used in all variables when there is not enough information to make the relevant inference, which is different from absence of the trait of interest. The following is a complete description of the coding scheme used in this project.

1. **Area:** This variable follows eHRAF’s classification system.
2. **Society:** This variable follows eHRAF’s classification system.
3. **OWC Code:** The Outline of World Cultures (OWC) is a classification system of the cultures of the world, which is used by eHRAF to organize its databases. It is used here as a unique identifier to merge the geographic coordinate data from D-PLACE for the purpose of mapping.
4. **Subsistence:** This variable follows eHRAF’s classification system, including Hunter-gatherer, primarily hunter-gatherer, horticulturalist, agro-pastoralist, intensive agriculturalist, other subsistence combinations. The category of commercial economy is excluded.
5. **Year of observation:** The years when the given society was observed by the ethnographers. This information is readily available under the name “field date” in the meta information of eHRAF World Cultures.
6. **Location:** The location where the learning event happens according to the ethnographer’s description.
7. **Location (type):** 1) indoor: learning happens inside a building; 2) outdoor: learning happens outside the buildings.
8. **Age of learner:** The learner’s age according to the ethnographer’s description.
9. **Age group of learner:** 1) Infancy and Early Childhood (0-5 years old), 2) Middle Childhood (6-12 years old), 3) Adolescence (13-16 years old), 4) Adulthood (>16 years old). For analytical purpose, this category does not concern the cultural differences on the perception of

developmental differences. When an age range is provided, the minimal age will be used in the classification of age group.

10. **Number of learner:** The number of learner according to the ethnographer's description.
11. **Age of model:** The model's age according to the ethnographer's description.
12. **Age group of model:** 1) *Infancy and Early Childhood* (0-5 years old), 2) *Middle Childhood* (6-12 years old), 3) *Adolescence* (13-16 years old), 4) *Adulthood* (>16 years old).
13. **Number of model:** The number of model according to the ethnographer's description.
14. **Transmission modes:** 1) *horizontal transmission*: social learning from individuals of the same generation, age group, or cohort, roughly within 4–5 years of age; 2) *vertical transmission*: children learning from their parents; 3) *oblique transmission*: social learning between individuals of distinct generations or age groups typically from an older generation to a younger generation. It can happen within or beyond extended kin group, such as learning from grandparents, uncles, or other adults in the local population; 4) *mixed*: different transmission modes occurred together in the learning process of a given technology.
15. **Social relationship:** The relationship between the learner and the model according to the ethnographer's description. For example, mother-son/father-daughter/etc.
16. **Transmission bias:** This category follows the framework developed by Kendal et al. (2018). 1) *kin-based bias*: learning a cultural trait from people who are your kin members, 2) *success-based bias*: learning a cultural trait from people because they are successful in this technology, 3) *prestige-based bias*: learning a cultural trait from people because of their high social status; 4) *conformist bias*: learning a specific type of knowledge from other people because it is displayed by the majority of local population; 5) *mixed*: different transmission biases occurred together in the learning process of a given technology.
17. **Learning process:** The learning process according to the ethnographer's description, such as “teach”, “imitate”, “observe”, etc.
18. **Learning process type:** 1) *individual learning*: Individual exhibits repeated attempts to learn a skill or develop new skills or knowledge on his or her own, including trial and error and individual practice; 2) *observational learning*: learner directly observes the actions of

models and attempts to replicate the observed actions, and no direct interaction between the model and learner is observed or mentioned; 3) *interactive learning*: this category features the verbal, gestural and bodily interactions between models and learners, be it formal (“teaching”) or informal (“group play”). It should be noted that these three categories are not mutually exclusive. On the contrary, these three categories are inclusive of each other, where interactive learning is built on both observation and individual practice, and observational learning involves individual practice.

19. **Time spent:** The time spent on learning according to the ethnographer’s description.
20. **Technology:** The subject of learning according to the ethnographer’s description.
21. **Technology type:** The types of technology were determined post hoc from collected ethnographic texts based on their frequency, including agriculture equipment making, bag making, basket making, boat making, clothes making, fishing devices making, general toolmaking, hammock making horse harnesses making, mat making, metal working, miscellaneous, musical instrument making, net making, ornament making, pottery making, raw material processing, ritual item making, rope making, stone working, toy making, trap making, vehicles making, vessel making, weapon making, weaving, woodworking. These categories are not mutually exclusive, and some categories are single out either because of cultural significance or the lack of relevant information. For example, weaving is a commonly documented skill that can be used in the manufacture of clothes, bag, basket, hammock, mat, etc, but it has its own category because no other contextual information is available to differentiate between them. Miscellaneous is used when the frequency of a specific toolmaking skill is lower than three. General toolmaking is an umbrella terms for phrasing like “utensils” or “tools” in the original text.
22. **Raw material:** The raw materials were also determined post hoc from collected ethnographic texts based on their frequency, including stone, clay, plant fiber, wood, bones, leather, composite, etc.
23. **Note:** This records interesting information that is not systematically documented in other variables, such as the difficulty of a skill as perceived by local people and ethnographers, access to raw material, and the restriction on the social status involved in the learning such as gender, wealth, religious status, etc.

24. **Reference:** This provides the relevant references, which can be linked to the relevant paragraph in the appendix.

2.3 Statistical analysis and visualization

3 Results

3.1 Descriptive results

3.1.1 What

Figure 2 displays the frequency distribution of various tool-making skills documented in non-industrial societies. The data reveal significant variation in the representation of different craft activities. Clothes-making emerges as the most frequently recorded skill ($n = 64$), followed by basket-making ($n = 49$), weapon-making ($n = 44$), and pottery-making ($n = 38$). Weaving ($n = 28$) and woodworking ($n = 26$) are also prominently represented. In contrast, certain specialized skills, such as vehicle-making and agricultural equipment-making, appear infrequently ($n = 3$ each).

This distribution suggests that daily utilitarian crafts, particularly those related to clothing and storage, are more commonly documented, possibly reflecting their widespread necessity and frequent practice. In contrast, more specialized or context-specific crafts, such as ritual item-making ($n = 6$) and horse harness-making ($n = 4$), are less commonly reported. The variation in frequency may reflect both the cultural importance of certain skills and the ethnographic focus on more pervasive or visible activities within these societies.

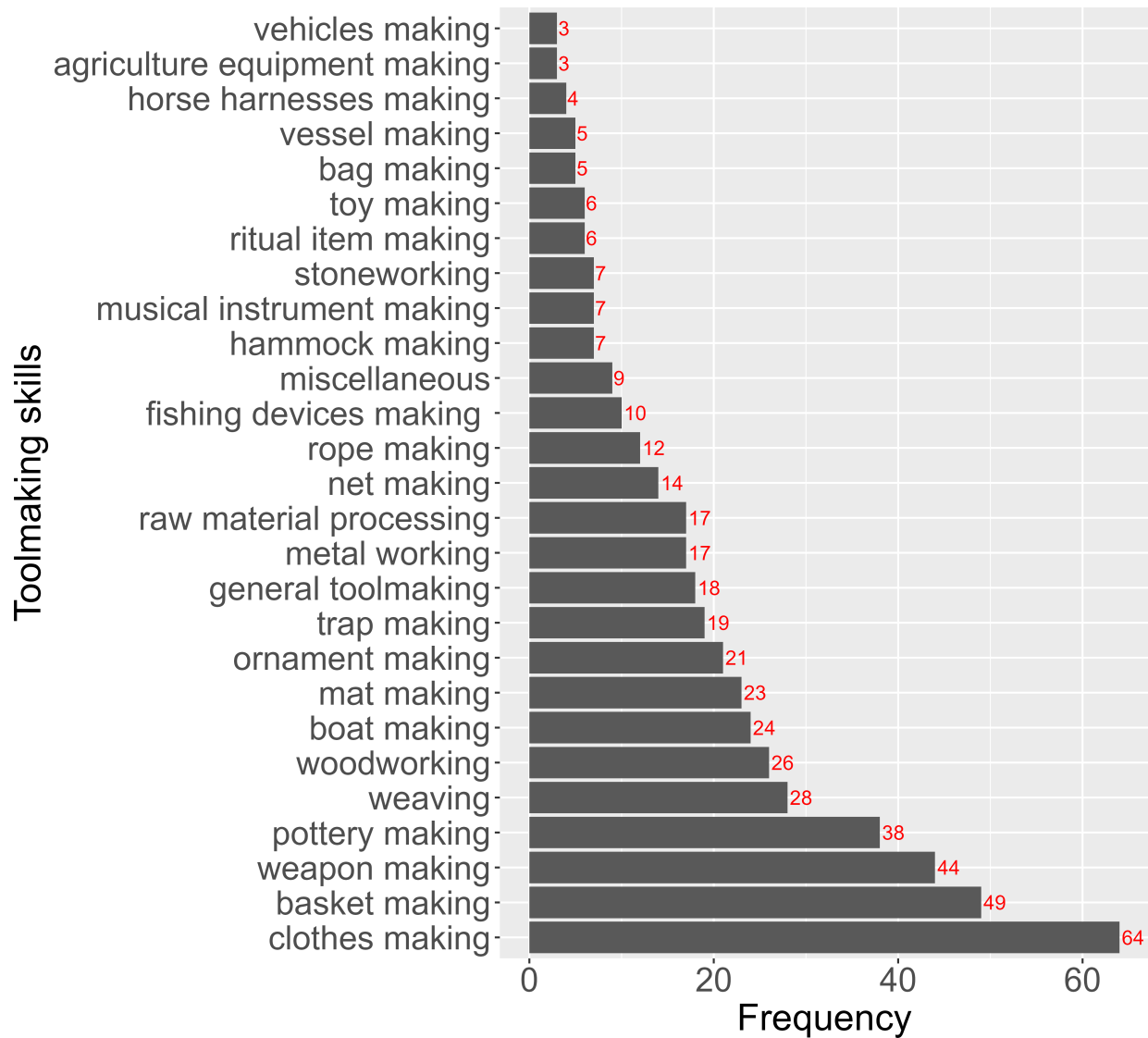


Figure 2: The frequency of toolmaking skills documented in ethnographic texts.

3.1.2 Who

Here the author analyzed the distribution of different modes of cultural transmission (**Figure 3**). The results indicate that vertical transmission, where skills are passed from parent to child, is the most frequently documented mode ($n = 162$). This is followed by vertical/oblique transmission ($n = 112$), where learning occurs both from parents and other adults within the community. Purely oblique transmission—learning from non-parental adults—accounts for 59 instances. Horizontal transmission, involving learning from peers, appears less frequently ($n = 10$), and mixed modes combining horizontal and oblique transmission are even rarer (e.g., hori-

zontal/oblique: $n = 6$; oblique/horizontal: $n = 1$). Additionally, some cases report a combination of all modes ($n = 11$). Notably, 118 instances lack specific information on transmission mode, highlighting gaps in ethnographic reporting.

These results suggest a strong reliance on kin-based learning pathway, particularly vertical transmission, for toolmaking skills in non-industrial societies. Taking a closer look at the social relationship between models and learners (see supplementary table 1), the author also found that the majority cases under the categories of oblique transmission involve grandparents, aunts, or uncles, which are also close kin members of the learner. This pattern may reflect the importance of familial and inter-generational knowledge transfer in maintaining craft traditions. Interestingly, the data show limited evidence for peer-to-peer (horizontal) transmission as contrary to what has been reported by Lew-Levy and her colleagues ([Lew-Levy, Kissler, et al., 2020](#); [Lew-Levy et al., 2023](#); [Lew-Levy & Amir, 2024](#)). It is possible that peer play activities or enculturation through interactions with older siblings falling into a similar age range are not that vital in the domain of toolmaking skills as compared with knowledge domains such as social and gender norms ([Lew-Levy et al., 2018](#)), considering the fact that several skills either require many years' practice to master ([Bock, 2002](#); [Gurven & Kaplan, 2006](#); [Kaplan et al., 2003](#)) and/or a certain degree of physical strength and motor control that is only available during or after adolescence ([Bird & Bliege Bird, 2002](#); [Bliege Bird & Bird, 2002](#); [Jones & Marlowe, 2002](#)).

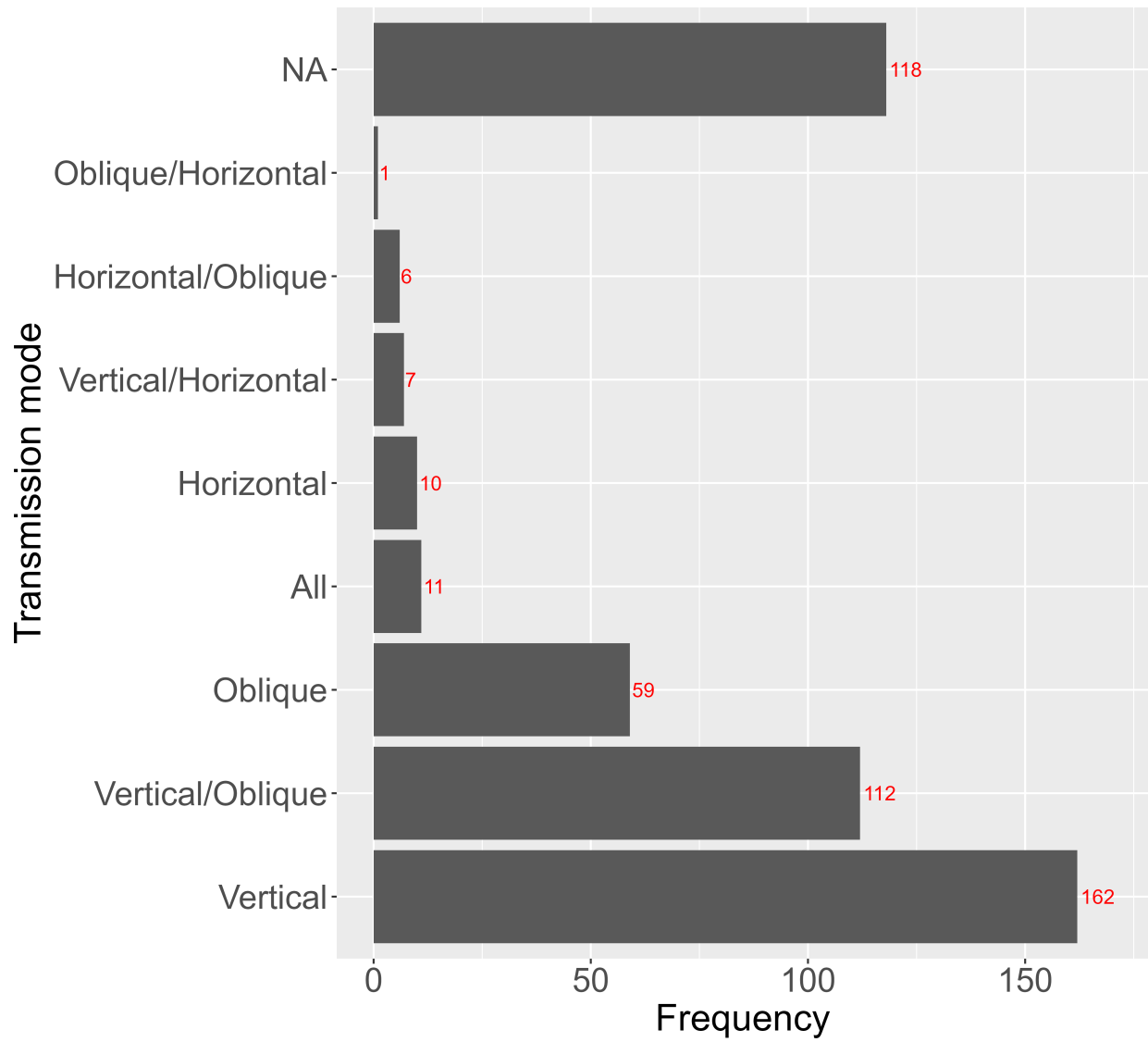


Figure 3: Transmission modes of toolmaking skills documented in ethnographic texts.

3.1.3 Where

As displayed in (Figure 4A), the majority of ethnographic cases ($n = 421$) lacked explicit information regarding the learning locations of toolmaking skills. Among the cases with documented locations, learning occurred most frequently indoors ($n = 38$), followed by outdoor settings ($n = 25$), with only two instances involving mixed indoor-outdoor contexts. Figure 4B further illustrates the relationship between learning environments and specific tool-making activities using a treemap. Indoor learning environments are predominantly associated with activities such as clothes making ($n = 6$), basket making ($n = 5$), and metal working ($n = 5$). In contrast, outdoor

learning environments are primarily linked to boat making (n = 4), trap making (n = 3), and net making (n = 2). Only two activities, metal working and stoneworking, were reported in mixed learning environments.

This pattern suggests that the physical setting of skill transmission is potentially tied to the nature of the craft. Activities involving fine, detailed work or those requiring specialized tools and materials are more often learned indoors, whereas tasks demanding larger spaces or environmental resources, such as boat or trap making, are typically learned outdoors. However, this distribution might simply reflect observers' biases and not necessarily the . The predominance of unreported spatial contexts of learning highlights the need for more systematic ethnographic documentation of learning environments in future research.

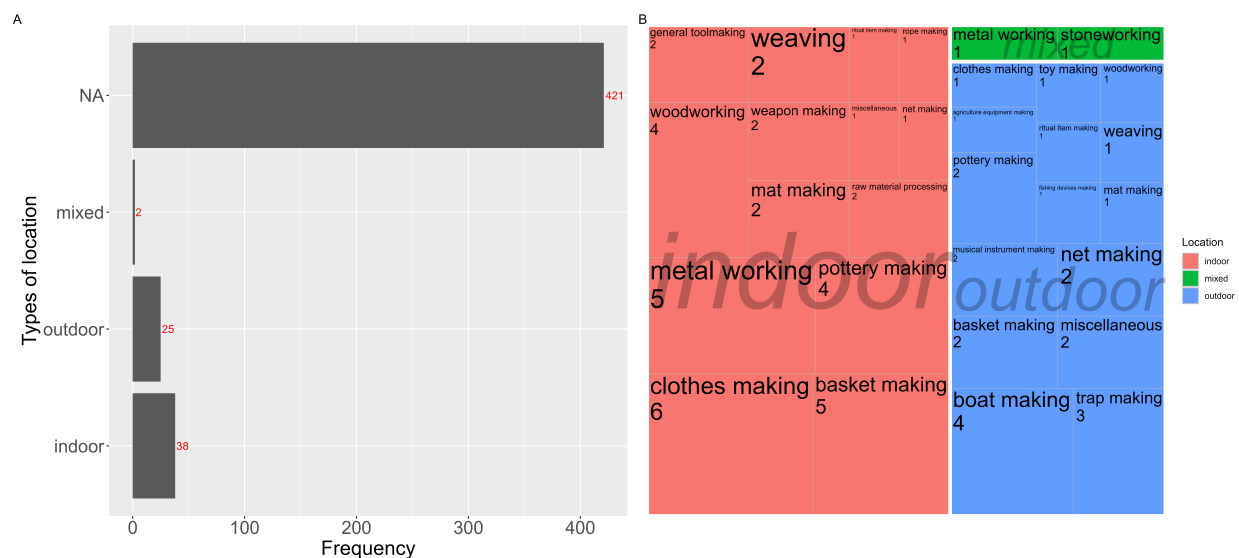


Figure 4: Location of toolmaking skill acquisition documented in ethnographic texts. A displays the frequency of learning locations. B is a treemap demonstrating the connections between craft categories and location locations.

3.1.4 When

The analysis of toolmaking skill learners' developmental contexts (**Figure 5**) revealed that most learning instances (n=274) did not have specific age-related information documented. Among the cases where age was specified, middle childhood (ages 6-12) showed the highest frequency of learning episodes (n=128), followed by adolescence (ages 13-16) with 47 instances. Infancy and early childhood (ages 0-5) accounted for 34 documented cases, while adulthood showed the lowest frequency with only 3 recorded instances. This distribution suggests that the transmission of

tool-making knowledge in non-industrial societies occurs primarily during middle childhood, a critical period in developmental psychology featuring multiple cognitive milestones such as advanced episodic memory (Ghetti & Bunge, 2012), theory of mind (Osterhaus & Koerber, 2021), and executive function (Wilson et al., 2018). Again, the substantial number of unspecified age cases (NA) indicates potential limitations in the ethnographic documentation of age-specific learning processes.

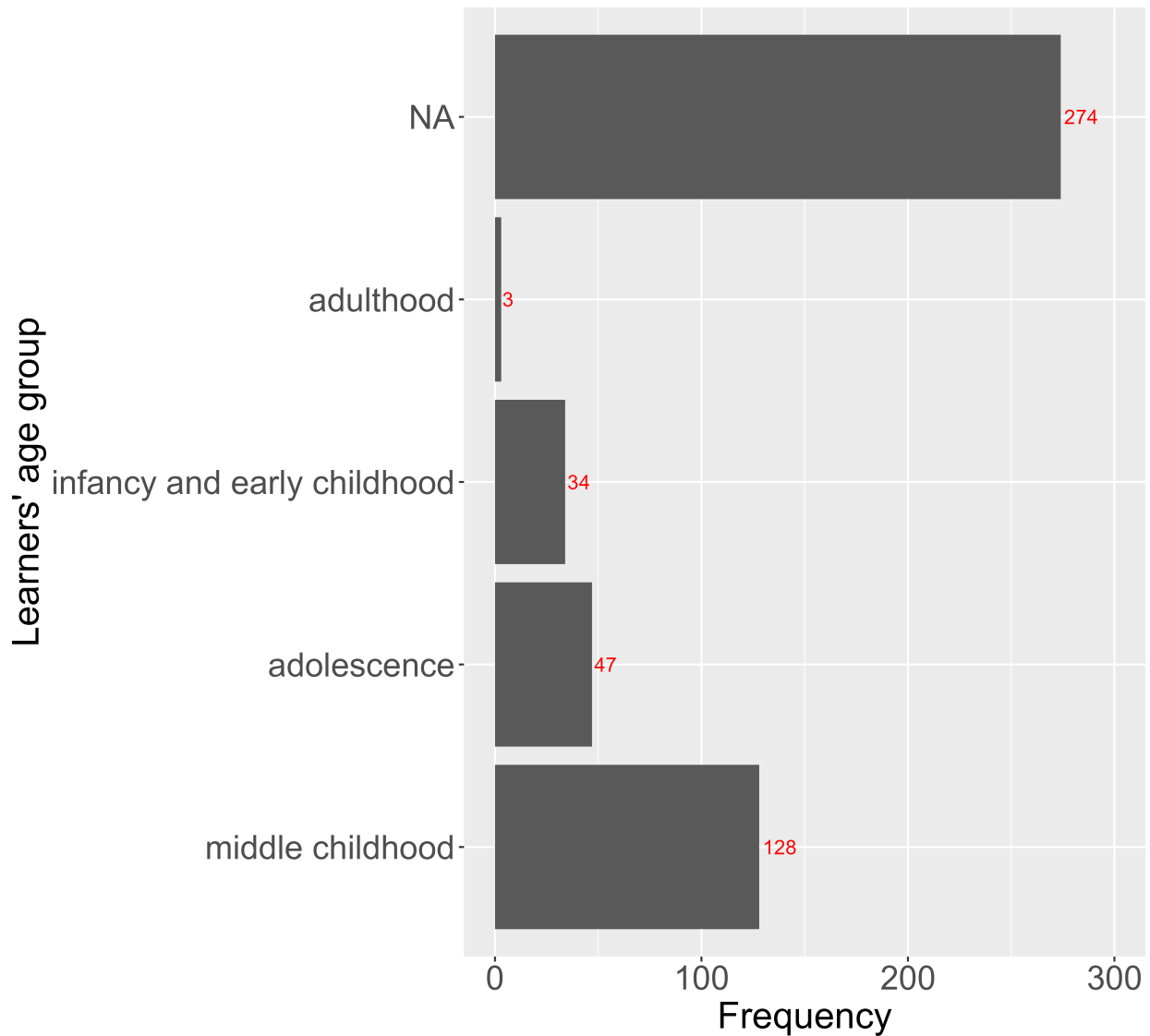


Figure 5: Developmental contexts of toolmaking skill acquisition documented in ethnographic texts.

3.1.5 How

Figure 6 presents the distribution of learning processes for toolmaking skills as . Interactive learning, which involves direct engagement between teacher and learner, was the most common method documented with 304 instances. Observational learning, where individuals learned by watching others without direct interaction, occurred in 62 cases. Individual learning, characterized by self-directed experimentation and practice, was recorded in 16 instances. In 104 cases, the specific learning process was not identified (NA). These results indicate that the transmission of tool-making knowledge in non-industrial societies relies heavily on social interaction, be it verbal or gestural, with observation serving as a secondary but significant learning method. The relatively low frequency of individual learning confirms the observer's bias towards social activities as opposed to following individuals as mentioned in the introduction, which by no means suggested that individual trial-and-error learning is not important in the acquisition of toolmaking skills . Rather, given the inclusive nature between three categories in this coding scheme, this result suggests that practice is integral to the learning process but pure individual practice without social interaction is rarely documented.

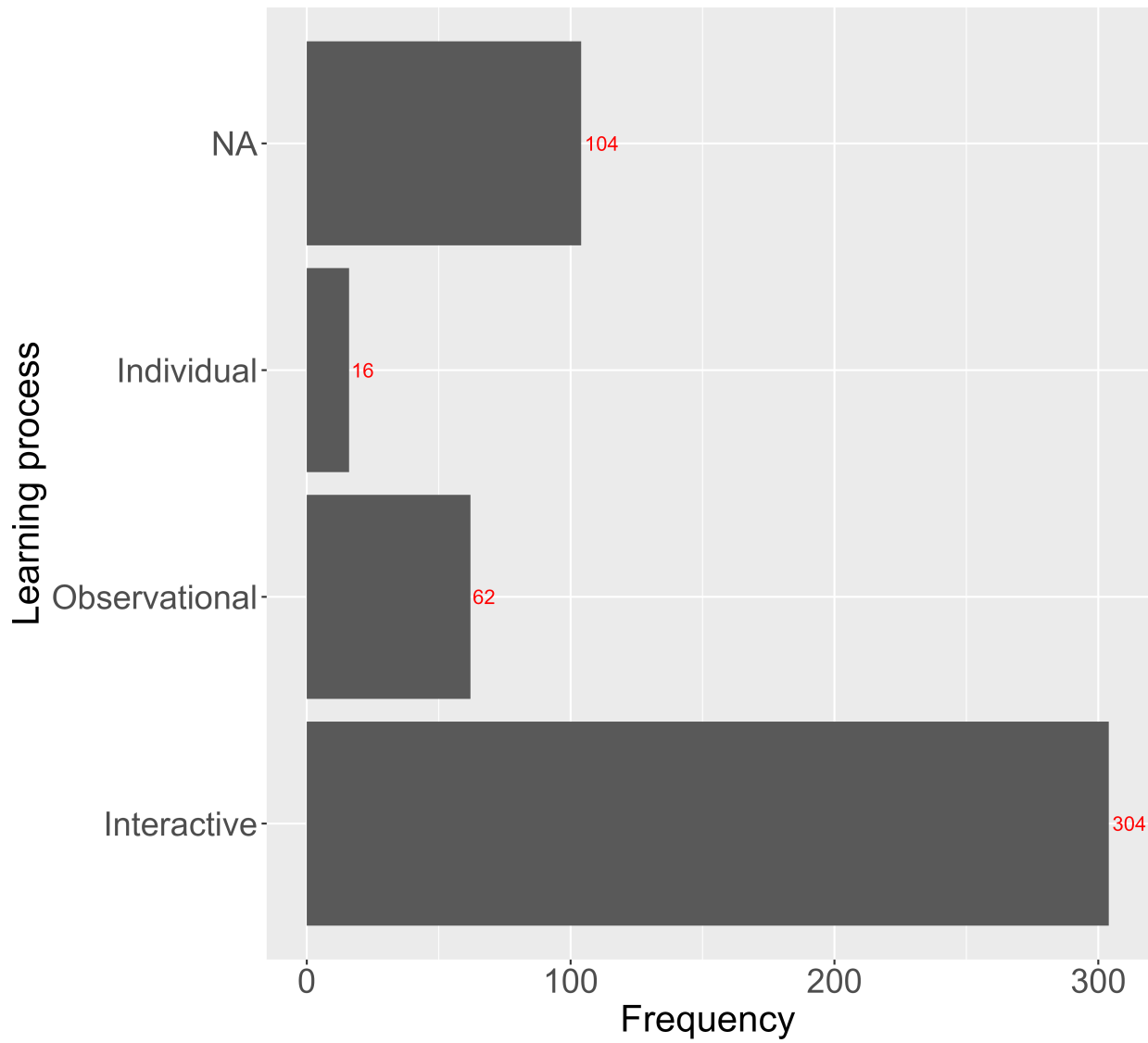


Figure 6: Learning processes of toolmaking skill acquisition documented in ethnographic texts.

3.1.6 Why

The distribution of transmission bias (**Figure 7**) agree with the result of transmission mode analysis as illustrated in the who section above. Kin-based transmission emerged as the predominant pattern with 288 documented cases, suggesting that tool-making skills are most often passed down within family units. Mixed transmission strategies, which combine different learning approaches, were observed in 33 cases, while success-based transmission occurred in 28 instances. Prestige-based transmission, where learners chose models based on their social status, was documented in 8 cases, and familiarity-based transmission appeared in only 3 instances. A

substantial number of cases ($n=126$) did not specify the transmission bias (NA). Again, this distribution shows that kinship plays a central role in the cultural transmission of tool-making skills, while other transmission biases occur less frequently. This finding is particularly interesting because traditionally the field of cultural evolutionary model focuses mainly on success/prestige bias or conformist bias (copying the majority), while kin-based bias remains understudied. A potential explanation is that success/prestige bias ([Chellappoo, 2021](#); [Chudek et al., 2012](#); [Jiménez & Mesoudi, 2019](#)) or conformist bias ([Henrich & Boyd, 1998](#); [McGuigan et al., 2012](#); [Wakano & Aoki, 2007](#)) may be more important in other cultural domains like the transmission of beliefs or ascetic preferences, while toolmaking usually is a long-term learning process that featuring more intimate social relationship.

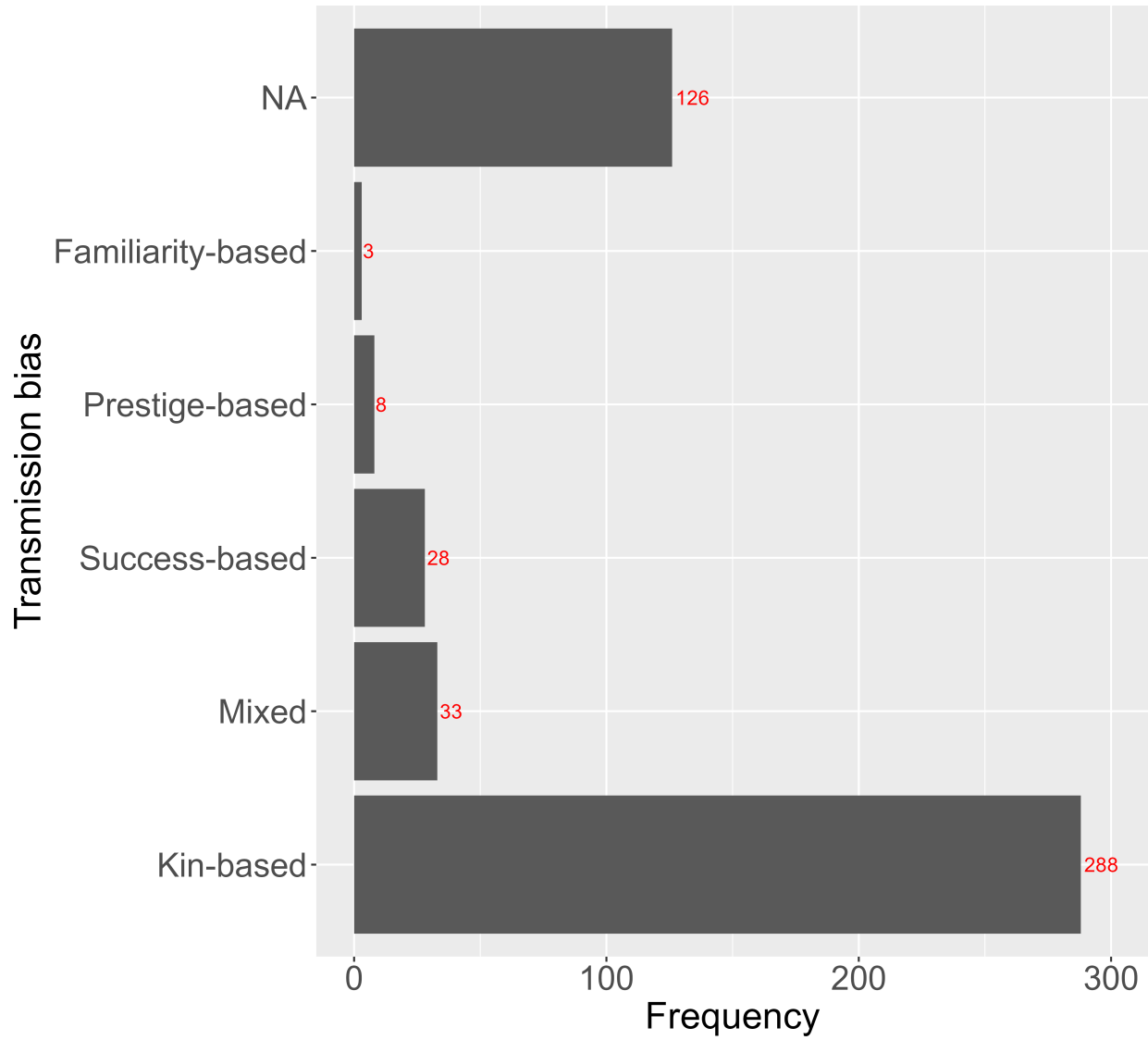


Figure 7: Transmission biases of toolmaking skill acquisition documented in ethnographic texts.

4 Qualitative insights

4.1 Protracted and staged learning trajectories

This study partly confirms Dallos’s (2021) sharp observation on the significant role of protracted learning in the formation of human culture, as several HRAF ethnographic sources suggested that the learning of toolmaking skills spans multiple developmental periods in an almost structural manner, where each periods feature different approaches to learning and varied task priorities. This applies to scenarios where people learn different different domains of technology

at different ages as well as people acquiring one toolmaking skill learn various technological components in a developmentally scaffolding manner. For instance, in Lancy's (1996: 161-162) classical ethnography on Kpelle children development, he noted that:

“Children learn three types of tasks at three different ages. They begin performing such rudimentary tasks of rice farming as cooking, pounding rice, and chasing away birds at a very early age, usually before they are 9. More difficult tasks, such as hoeing on the farm, net weaving, and trap making, are begun around age 9 and mastered by age 13. Rarer, more complex skills such as cloth weaving, bag weaving, blacksmithing, and medicine making are acquired in the late teens and may be learned after marriage.”

Canoe making, a skill that demands not only the acquiring of technical know-how but also the development of proper physical strength to handle heavy logs, best illustrates the case of long training period required for a single toolmaking skill. For example, Wilbert (1976: 360) commented that among the Warao hunter-gatherers' skill training for men last until 25 years old and he further (Wilbert, 1977: 23) wrote that:

“A boy is associated with canoe-making from early childhood on, initially as spectator, when he accompanies his father and the other men of the family to the forest and watches the production process. Later, as an adolescent, he picks up the axe himself and participates in the process according to his increasing dexterity. By the time he gets married, a young man commands the basic skills and may even have a small canoe to call his own.”

This pattern of prolonged and staged learning has also recurrently occurred in other ethnographic observations on other indigenous communities such as Chuuk/Truk (Goodenough, 1951: 53-55) in Micronesia, as well as Kapauku (Pospisil, 1958: 40-41) and Trobriands (Austen, 1945: 194-195; Scoditti, 1989: 56) in Melanesia (e.g., “15-20 years of apprenticeship”). However, given the inherent constraint that most ethnographic sources used here only provided a fragmented narrative on craft learning, it is still unclear how common is this mode of protracted learning in the manufacturing of smaller-sized tools that does not necessarily require high

level of physical strength as debated in several empirical studies regarding humans' unique life history pattern (Bliege Bird & Bird, 2002; Jones & Marlowe, 2002).

Initiation ceremony is a period for intensive toolmaking skill training, along with other skills required for adulthood: Laguru, Mende, Eyak, Kaska, Pomo, Maleluka, Tukano, Goajiro, Ona. For example, Hamdani (Hamdani, 2001: 11-12) described the female initiation rites in Luguru, East Africa.

“The ‘mwali’ initiation rite... She explained that she literally lived in this hut for a period of three years... It was her mother who trained her to sort out green vegetables, to pound the maize and to prepare food. It was her aunts who taught her how to make sleeping mats, baskets, pottery and even caps. Mama Fulora explained that during her grandmother’s times, girls used to learn how to make clothing from barks of trees.”

Similarly, Hugh-Jones (1979: 86-87) provides a detailed narrative regarding the importance of initiation ceremony in toolmaking skill acquisition with the example Tukano people.

“For the initiates, the marginal period is a time of education... Then the elders systematically teach the initiates to make all the different kinds of basketry (carrying baskets, sieves, flat trays, cassava squeezers, etc.)... In addition to basketry, the initiates are taught how to make the feather head-dresses and other ritual ornaments used in dances... Just before the blowing of pepper, the initiates are taught to weave conical fish traps which they go and set in streams in the forest, accompanied by their masori.”

4.2 The distribution of expertise

In many non-industrial societies, while basic toolmaking skills are widely shared, individuals recognized as experts are relatively rare. This distribution of expertise can be attributed to two key factors. First, the role of innate talent or individual aptitude is often emphasized, as seen among the Nuba, Bemba, Yukaghir, Malay, and Southern Coast Salish, where certain individuals are believed to possess natural abilities that set them apart in craftsmanship. Second, access to specialized resources can limit the development of expertise. For example, among the Malays,

access to high-quality materials and tools for silversmithing is restricted, preventing widespread mastery of the craft. These dynamics have significant implications for archaeologists aiming to identify varying skill levels in the archaeological record. Artifacts may reflect both general community knowledge and the specialized work of experts, making it essential to consider factors like resource availability and individual skill variability when interpreting production quality and technological complexity in past societies.

Nuba ([Nadel, 1947](#): 71-72)

“Rather, craftsmanship appears irregularly, as a result of the inclinations and the cleverness of certain individuals which, once they become known, are utilized by others who are not so ‘clever with their fingers’. Everybody can, in theory, make baskets or beer-strainers; but in most groups we find ‘experts’ to whom people would go for these things in preference to making them (much less satisfactorily) themselves.”

Nubian ([Kennedy, 1977](#): 25)

“Nubian building techniques were passed down through families from generation to generation, but a few men were regarded as specialists. Some of these were responsible for local carpentry needs—for sakkia repair, the manufacture of beds, wooden chairs, donkey saddles, and the like; some were masons.”

4.3 Causal knowledge in the craftsmanship

Teaching and learning almost always appear together in modern education contexts. However, these two concepts are not exactly symmetrical in the sense that learning has a much broader connotation and is widely believed to present in an extensive range of species in the animal kingdom, while teaching is not. Or put it differently, learning does not necessarily require the presence of a teacher, while teaching, in its narrowest sense, implies the existence of a learner, even if it is just an imaginative one. In fact, depending on the definition one adopts, it has been long debated that whether teaching is a cross-cultural or even cross-species phenomenon (Boyette and Hewlett, 2018). Generally speaking, there have been three competing ideas around the presence and scope of teaching. The first position is represented by classical cultural and psychological

anthropologists like Margret Mead (1964), arguing that teaching does not exist or is rare in small-scale cultures. More specifically, Mead characterized small-scale societies as “learning cultures” where there is a general consensus on what was important to learn and the learners, instead of teachers, initiated the acquisition of skills or knowledge and acquired them relatively easily. This is in sharp contrast with what Mead addressed as “civilized societies”, which are dominated by “teaching cultures”. In these societies, it is the teacher that determines what should be taught to the younger generation and how they are taught. Mead also claimed that Western political and religious institutions dedicated to conversion, indoctrination and colonization are the roots of its modern teaching and educational systems. This position was later inherited and developed by Lancy (2010: 97), who provided an operational definition that teacher is “nuanced, student-centered, developmentally appropriate instruction by dedicated adults” and is a recent product of “a long process of education changes”. Lancy further suggested that even the idea of parents as teachers emerged rather late. It is the multiple demographic transitions characterized by smaller birth cohorts and drastically decreased infant mortality that eventually allowed parents to spend more time and energy to interact with and educate offspring to fit personal ideals. Despite bearing a clear Western-center idea of teaching, this position was justified by some extensive and ecological valid field observations.

The middle position is typical of developmental and comparative psychologists, among whom one of the most representative figures would be American psychologist Michael Tomasello. Tomasello and his colleagues believed that teaching is unique to humans and part of human nature and defined it as “a behavior in which one animal intends that another learn some skill or acquire some bit of information or knowledge that it did not have previously” (Kruger et al., 1996: 374). This relatively inclusive approach implicitly affirms hunter-gatherers’ ability to teach, which is also characterized by the extensive use of the WEIRD population in controlled experiments. What is really implied in this definition is the possession of false beliefs, namely an individual needs to understand that one’s own belief might be different from others’ beliefs, especially when these beliefs are conflicting. False belief understanding has long been regarded as the richest signature of the cognitive capacity famously known as the theory of mind, which is the ability to infer the mental states of other individuals. Another cognitive prerequisite of teaching according to Tomasello would be metacognition, namely the awareness of one’s own knowledge in different domains. Human children usually developed the cognitive abilities of the theory of mind and metacognition between 3 and 5. Accordingly, A series of empirical

studies summarized in Strauss et al. (2002) showed that children as young as age 3.5 years, who have never been taught to teach, engage in teaching through demonstrating games rule to other children and helping them learn how to play. In the late 1990s and early 2000s, it was still widely believed that these are uniquely human cognitive traits, especially the understanding of false beliefs, however, the experimental studies conducted during the past two decades have fundamentally subverted this idea as multiple groups of researchers have shown that chimpanzees can pass the false-belief tests (e.g., Kano et al., 2019).

This new finding regarding non-human primates' ability to understand false beliefs naturally introduces the third position that teaching exists among human and non-human animals. Specifically, Caro and Hauser (1992: 153) argued that "an individual actor A can be said to teach if it modifies its behavior only in the presence of a naive observer, B, at some cost or at least without obtaining an immediate benefit for itself. A's behavior thereby encourages or punishes B's behavior, or provides B with experience, or sets an example for B. As a result, B acquires knowledge or learns a skill earlier in life or more rapidly or efficiently than it might otherwise do, or that it would not learn at all." Their definition provides a phylogenetically productive method that allows us to investigate the processes through which some critical cognitive phenotypes belonging to humans evolved, although the underlying neuroanatomical and cognitive bases still remain elusive to a large extent. One of the most convincing examples of non-animal teaching under this definition would be wild meerkats. Thornton and McAuliffe (2006) showed that older meerkats will kill or disable scorpions (one of their major food sources) in front of junior meerkats and monitor the behaviors of the latter. If the juniors did not attempt to handle the prey, the older ones will even use their nose or paws to nudge the prey repeatedly until it was handled and consumed by juniors. With the increasing ages of junior meerkats, the frequency of killing or disabling prey in the presence of them decreases significantly. This behavior incurred additional costs for the older ones as the prey might escape.

As far as I am concerned, the debate is analogous to the question of whether chopsticks should be counted as eating utensils given their dissimilarity with forks and knives in form and texture. The minimality and flexibility in conceptualization usually lead to a more productive and inspiring research line. To reach a balance between productivity and falsifiability as reflected in these different stances, Hewlett and Roulette (2016) took a minimalist approach and defined teaching as "an individual modifies her/his behaviour to enhance learning in another", which is also the

position held by the author. In particular, they cautioned that teaching cannot be a by-product of another activity. For example, one individual warns another about an incoming danger, in which the latter individual may learn some new information, but it should be not counted as teaching since there is no behavioral modification presented by the former individual with the intention of enhancing others' learning activities. Under this definition, teaching was identified through video recordings during as early as the infancy period in a variety of forms among the Aka hunter-gatherers of the Central African Republic and Republic of Congo, ranging from natural pedagogy to opportunity scaffolding to verbal instruction. For instance, it was observed that caregivers sometimes directly demonstrated how to use a tool (knife / digging stick / machete) in front of babies and waited for a few seconds and then demonstrated the object again. Adults also taught infants by moving or redirecting them to another area when they were observed to display some dangerous behaviors (almost falling out of a tree) or behaviors that are deemed to violate the social norm of the community such as slapping other persons. Another noteworthy finding of this research is that most teaching behaviors documented in the Aka foragers are usually very short and subtle, lasting only a few seconds but occurring repeatedly. The combination of short durations and observational subtlety of these types of teaching behaviors provide one possible explanation on why teaching was believed to be absent or rare for early anthropologists, in addition to the debates around the definition.

Explicit transmission of causal knowledge appears to be rare in non-industrial societies, aligning with longstanding debates about the nature and prevalence of teaching across cultures. If teaching is narrowly defined as the deliberate and structured transmission of explicit causal knowledge, then its scarcity supports earlier claims by scholars like Mead (1964) and Lancy (2010) that teaching is limited or absent in small-scale societies. In these contexts, learning often occurs through observation, imitation, and participation rather than through formal instruction. This perspective challenges Western-centric definitions of teaching that emphasize explicit explanation and structured guidance. My findings suggest that teaching in non-industrial societies may be subtle, brief, and embedded within daily activities, making it less visible to researchers employing narrow definitions. Therefore, recognizing the diversity of teaching practices requires adopting broader, more inclusive definitions that account for indirect and informal forms of knowledge transmission.

Additionally, these findings contribute to ongoing discussions in cultural evolution research,

where experimental studies have shown that explicit causal understanding is not necessary for the transmission of complex cultural traits. My observations support this argument, demonstrating that individuals in non-industrial societies can effectively learn and transmit toolmaking skills without formal instruction in causal mechanisms. This suggests that high-fidelity cultural transmission can occur through embodied practice and situated learning rather than explicit teaching. Such insights reinforce models in cultural evolution that emphasize imitation, emulation, and practice over deliberate explanation. Recognizing that causal knowledge is not a prerequisite for cultural continuity challenges assumptions about the role of explicit reasoning in technological advancement and highlights the importance of contextual, experience-based learning in sustaining complex cultural traditions.

And this has been recently brought up by Harris et al by ask Hadza skilled hunter to identify the causal relationship between specific bow design and their performance. Their performance is not very good. They vaguely know what kinds of shape of bow would be more effective in hunting, but this holistic and technical know is hard for them to tease out. It is also commonly mentioned in many paragraphs I wrote.

Kaska ([Honigmann & Bennett, 1949](#): 185-186)

“Education. Most education for living is received by the child in the course of performing routine activities. Such education is unformalized and undirected... Sometimes generalized instructions are furnished to children who, in carrying these out, often become confused. Thus we observed one fourteen year old girl make tea by putting the leaves into cold water. Nobody referred to the muddy quality of the beverage. The same kind of teaching was used with the ethnographer. Activities were illustrated but never explained in detail, the exposition of principles being ignored.”

4.4 Alternative pedagogies: Co-production and miniature model

Instead of providing explicitly step-by-step causal knowledge-oriented teaching, a common way of acquiring a toolmaking skill is that experienced model co-produce an object with novice. When novice made mistakes, the experienced model take it back and correct the mistake and then return to the novice again for them to finish. During this process, no or little verbal explanation regarding why a specific step is necessary or how to avoid similar mistakes in future is

provided, as vividly described in Lancy's (1975: 151) childhood-focused dissertation fieldwork in Kpelle horticulturalists in Western Africa.

“Wollokolle's youngest wife, Goma was an expert at Bii paa or bag weaving... Now she is teaching a teenaged girl of 16 how to do it. The girl has prepared the strands of palm leaf and brought them to Goma to help her get started. The two alternate at weaving the bag. The girl makes a mistake and Goma unravels what she has done, does it correctly for awhile, then lets the girl take over again. Goma can't explain how to do it, either to me or the girl. She can only show and guide our hands as I, too, make an attempt to learn.”

On top of this Kpelle bag weaving case, similar social interaction dynamics has also been observed among Garo people in South Asia regarding how the skill of basket making is transmitted between male model and learners (Burling, 1963: 116).

“Only after he is about twelve years old does anything like formal teaching take place. The most difficult skill to learn or to teach is basket-making, which the father or some other man must impart to a boy. The instruction consists mostly of telling a child to make a particular basket as he has watched others do and then reprimanding him if he does it wrong. Occasionally a man will demonstrate, but I never saw anyone who was able to analyze and point out the mistakes another was making. If a person makes a mistake in a complex basket, for instance, his instructor will take the basket away, redo it himself, and then tell him to do it that way henceforth. No effort is made to break the complex weaving process down into steps so as to make it easy to understand and learn systematically, although it is recognized that certain baskets are easier to weave than others. Fathers tell their sons to make the easier types first, and then gradually lead them to the more difficult ones.”

This recurring phenomenon, as also documented in Shona (Aschwanden, 1982: 62), Tallensi (Fortes, 1938 : 60-61), Tzeltal (Nash, 1970: 274-275), Gros Ventre (Flannery, 1953: 159-160), and many other societies, has some deep implications for archaeologists who are interested in identifying the skill level of producers based on certain morphological traits of individual specimen or a whole artifact assemblage. In this pedagogical scenario, the experienced individual and

novice co-produce an item, and part or all of the mistakes made by novices can be erased by the former's behaviors. The coexistence of fossilized actions of both stakeholders preserved in archaeological materials often provide us a precious opportunity to reconstruct the complex learning behaviors in the past, be it stone artifacts ([Assaf et al., 2016](#); [Assaf, 2019](#)) or pottery vessels ([Crown, 2007](#)).

Another commonly used pedagogical approach without resort to explicit causal knowledge transmission is the making of miniature models for and by children underage. To accommodate childrens' developmental constraint in hand size and physical strength, more experienced individual make smaller and often simplified version of the target artifact and then passed it to children as a model to begin with. It has been observed in many different societies and technological domains, including Blackfoot ([Ewers, 1958](#): 103) bow and arrow making, Trobriands ([Weiner, 1976](#): 192) net making, Jivaro ([Harner, 1973](#): 92-93) pottery vessels and figurines, Mbuti carrying basket making ([Turnbull, 1965](#): 125-126), etc. In terms of their implication for material culture research, these miniature models possess the interesting properties serving both as toys and tools ([Lew-Levy et al., 2022](#); [Riede et al., 2023](#)), and their producers can be either adults or children.

4.5 Task-dependent learning locations

The findings reveal that different toolmaking tasks are learned in distinct contexts, with raw material procurement typically occurring in quarries and finishing tasks taking place in residential areas. This task-dependent distribution of learning environments has important implications for archaeologists aiming to reconstruct learning processes from the archaeological record. It suggests that evidence of learning and skill transmission may be spatially segmented, with early-stage production debris concentrated at resource extraction sites and more refined production traces located in domestic settings. Archaeologists must therefore consider the spatial organization of artifact assemblages to identify stages of skill acquisition and social learning. Overlooking this spatial variability could lead to incomplete interpretations of how technical knowledge was transmitted. Recognizing the contextual nature of learning also underscores the need for integrated analyses of production sites and domestic spaces to fully capture the complexity of past learning environments.

Zapotec ([Cook, 1968](#): 189-191)

“In San Sebastian the finishing tasks in metate and mano production are taught in the residence lot; all preliminary and intermediate production tasks are taught and learned on location in the quarries. In San Juan, on the other hand, where the quarries are close to the village and more accessible than those in San Sebastian, metate size blocks of stone (trozos) are often carted from the quarry to the metatero’s residence lot in the village. Thus, the residence lot is, in many instances, the San Juan metatero’s workshop.”

4.6 Skill training for the community integration

Several sources in this study indicate that local communities perceive toolmaking not merely as a survival skill but as a vital process for social integration. This perspective aligns with Lave and Wenger’s (1991) concept of “communities of practice,” where learning is inherently social and occurs through active participation in shared cultural practices. In these contexts, engaging in toolmaking allows individuals, particularly younger members, to become integrated into the social fabric of their community. Through observation, imitation, and guided practice, learners not only acquire technical skills but also internalize social norms, values, and identities associated with craftsmanship. This participation fosters a sense of belonging and reinforces communal bonds, demonstrating that toolmaking serves both practical and social functions. Thus, tool production becomes a medium through which individuals transition from peripheral participation to full membership within their cultural group, highlighting the dual role of skill transmission in sustaining both material needs and social cohesion.

Tzeltal ([Nash, 1970](#): 321)

“Learning agricultural tasks and pottery-making in the household is part of adjusting to membership in the community. Parents are responsible for teaching the value of industriousness and are held accountable throughout the life of the child. Agricultural tasks are scheduled in a yearly cycle that provides sufficient flexibility for a man to carry out community and familial ceremonial responsibilities.”

Kpelle([Lancy, 1996](#): 154-155)

“In each of the previous cases, we see individuals acquiring weaving skills to enhance their own social participation. As Sua indicates, if a girl wants to “look nice,” to match the appearance of women, she must learn the skills to dress and decorate herself. In attempting to do so, she will be accorded the status of a “legitimate peripheral participant.” She may be supplied with materials and someone competent may agree to model the skill and also to critique the fledgling product. Miniature tools may be supplied.”

5 Conclusion

What: The three most frequent toolmaking skills are clothes making, basket making, and pottery making, documented in almost all non-industrial societies regarding of subsistence strategies or geographical distribution.

Who: People learn most toolmaking skills from their close kin members, particularly parents. Extended family members like grandparents and uncles/aunts are also heavily involved in this process.

When: For many skills, they started the learning processes in middle childhood (6-12 years old), which can last for several years. The initiation ceremonies provide opportunities for a period of intensive training.

How: Although individual practice and trial and error always exist, toolmaking skill acquisition involves lots of social interactions, during which causal knowledge is rarely explicitly transmitted. Instead, two commonly used pedagogical approaches are 1) object co-production between experienced and inexperienced individuals as well as 2) miniature models manufactured for and by underage novices.

Where: Indoor learning scenarios are more frequently documented compared with the outdoor ones. The former is often associated with task involving meticulous motor control and specialized tools and materials, while activities demanding larger spaces or environmental resources are typically learned outdoors. For complex crafts involving multiple steps like raw material procurement, the locations are dependent on the specific task involved.

Why: Toolmaking skill transmission is regarded as an important way of community integration.

I see it as a stepping stone instead of a definitive text for a better understanding of the subject matter, which merely serves as a starting point for an open dialogue and a project to be collaborated with future researchers like the “Play Object Play” database (Riede et al., 2023). This article also provides many empirical insights into some key debates or theories in anthropological discourses, such as the existence of teaching, community of practice, .

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